

LAB 4 · INTERCONNECTION NETWORKS

Collective Communication for Modern GPU Training

Router-level multicast, express-link bypass, and streaming tensor all-reduce in gem5 Garnet

SHARE

SHORTCUT

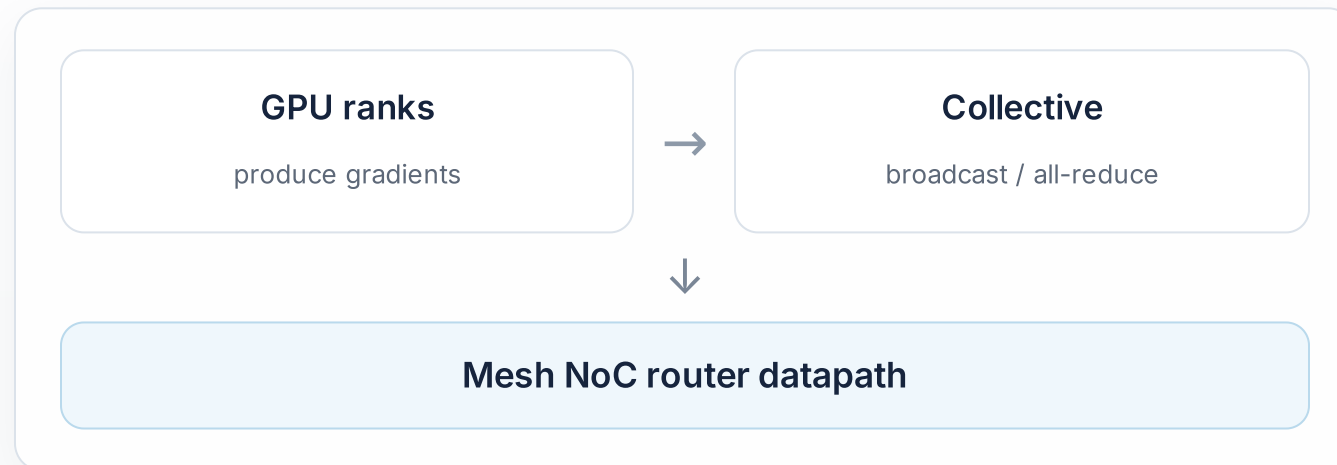
STREAM

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Topic 3 + Topic 4 extension

Collectives sit on the training critical path

Every training step alternates between local compute and a synchronization contract.



Share duplicated prefixes

Replicate a broadcast only where paths branch.

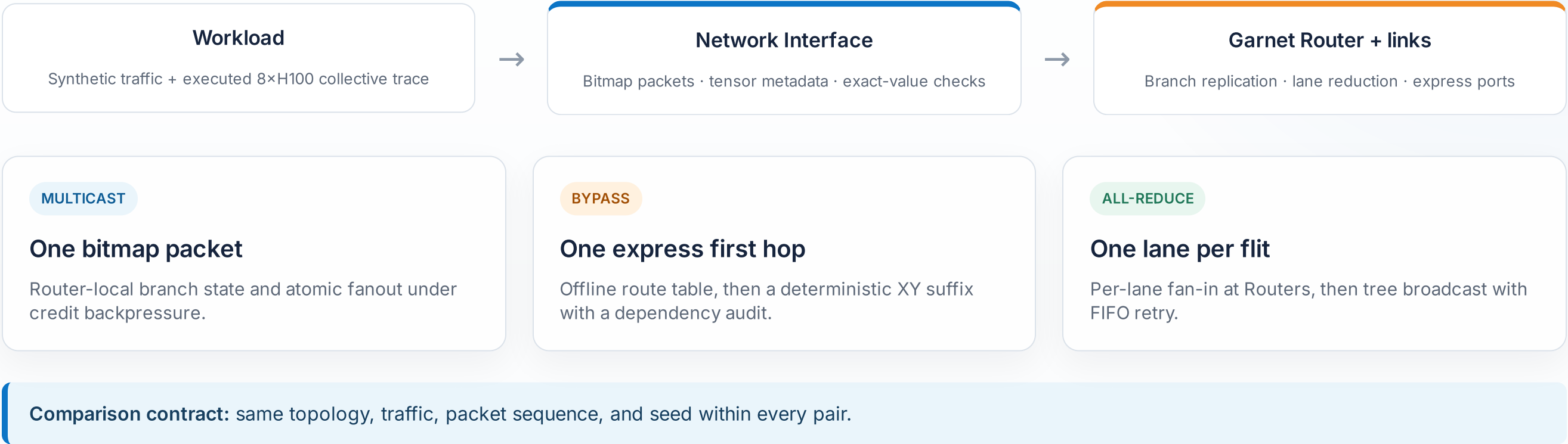
Shortcut long routes

Use physical express links only when wire cost is justified.

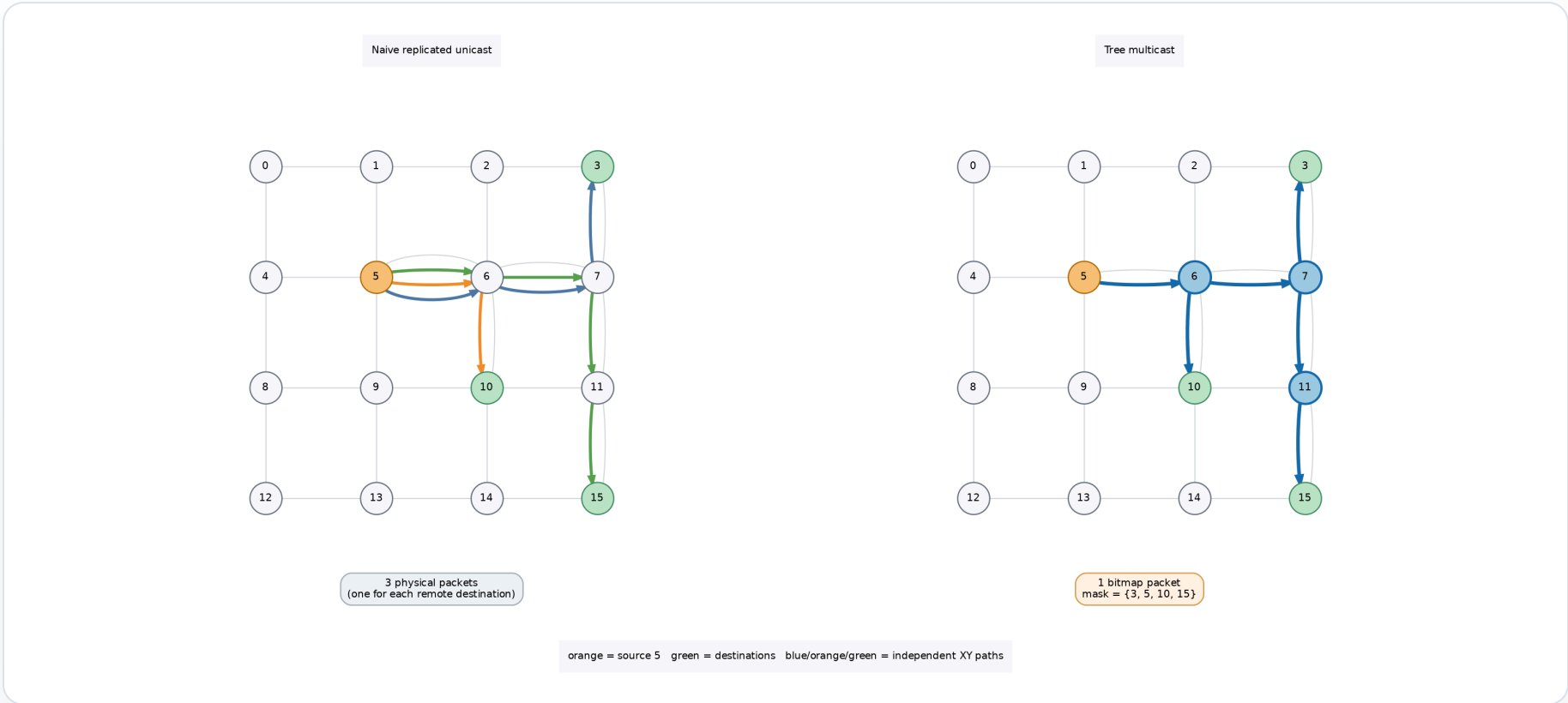
Stream tensor requests

Preserve multi-flit request boundaries instead of serializing lanes.

Three mechanisms, one auditable simulator



Share: replication moves into branch Routers



Source Router 5; destinations {3, 5, 10, 15} on a 4×4 Mesh.

Baseline

One physical packet per destination; common XY prefixes repeat.

Tree multicast

One 64-bit destination bitmap; copies appear only at selected branch Routers.

Correctness rule

HEAD/BODY/TAIL advance only when *all* selected branches have output VC credit.

Multicast reliably removes traffic—not every stall

39.51%

mean internal-link flit reduction

4.790×

mean packet-latency speedup

10 / 162

throughput regressions; worst −18.24%

Traffic saving grows with fanout

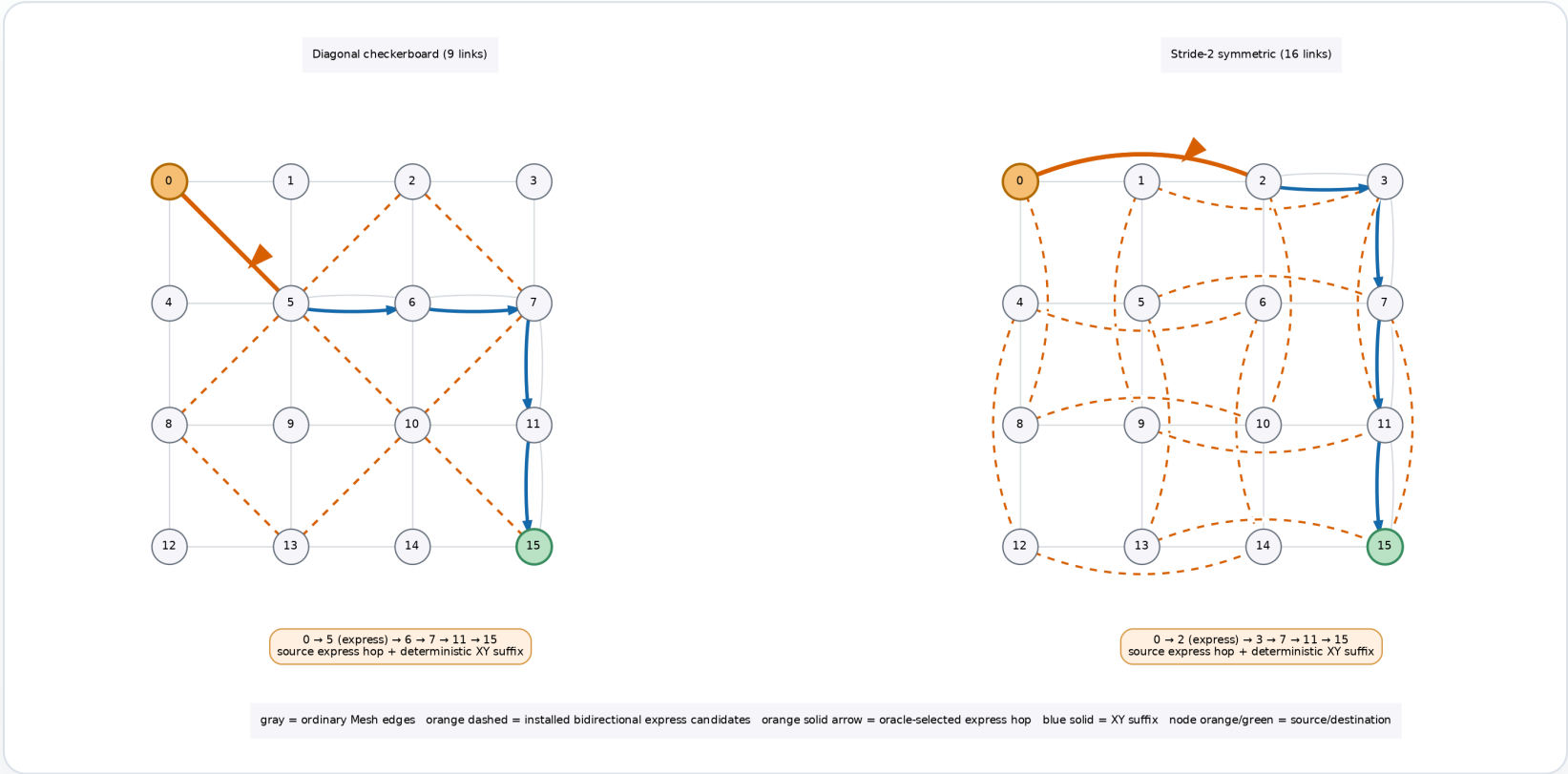


Why can less traffic still lose throughput?

Atomic fanout couples fast and blocked branches. All ten regressions occur in four-destination cases, concentrated in longer, loaded requests.

Matrix dimensions: 4×4 Mesh · fanout {4, 8, 16} · packet size {1, 4, 16 flits} · multiple traffic/load conditions.

Shortcut: add links, but make every route auditable



Orange: installed express candidates · Blue: deterministic XY suffix.

1 · Build candidates

Diagonal or stride physical links

↓

2 · Select one first hop

Only if remaining Manhattan distance shrinks

↓

3 · Audit dependencies

Accept only an acyclic channel-dependency graph

↓

4 · Follow XY suffix

Deterministic path after the shortcut

Wire cost decides whether fewer hops are valuable

Express design	Latency speedup	Throughput change	Aggregate traversal reduction
Diagonal · distance-scaled	1.629×	+6.41%	-2.82%*
Diagonal · optimistic	1.723×	+10.61%	-7.03%*
Stride · distance-scaled	0.981×	-1.94%	+15.50%
Stride · optimistic	1.103×	+3.68%	+10.10%

Diagonal wins latency

Short links survive distance-scaled timing and improve contention behavior.

Stride wins hop count

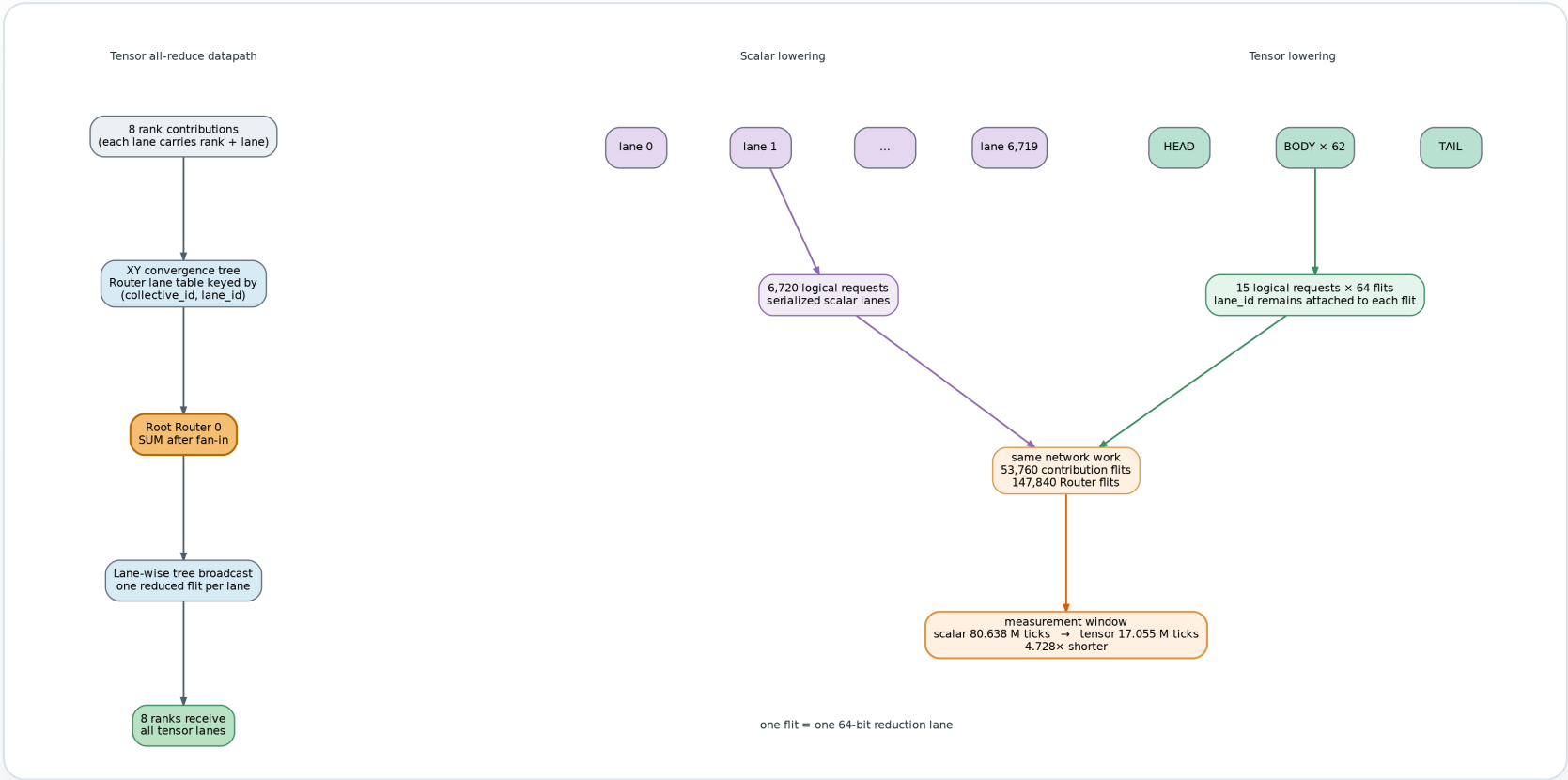
Longer links remove Router traversals, but wire latency and radix erase the gain.

Physical cost differs

4×4: diagonal adds 9 links (wire proxy 18); stride adds 16 (wire proxy 32).

*Aggregate traversals may rise when a faster design delivers more packets; read together with throughput. Each design: 1,536 paired cases across sizes, traffic patterns, packet lengths, loads, and seeds.

Stream: keep a tensor as one multi-flit request



Per-flit identity

(collective_id, lane_id) stays attached through reduction and broadcast.

Backpressure-safe state

Per-Router lane table + pending-forward FIFO retries when output credit is unavailable.

Exact completion

Every rank checks every lane value; residual lane/FIFO state fails the run.

Same network work, 4.728x shorter replay window

4.728x

measurement-window speedup

6,720

scalar requests

15

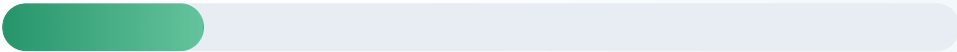
tensor requests

Scalar lanes



80.638 M ticks

Tensor stream



17.055 M ticks

Controlled comparison

Both lowerings perform 53,760 rank-contribution flits and 147,840 Router flits. The gain comes from request representation and completion scheduling—not less network work.

15 measurement-phase all-reduce events · 1/1024 byte/time scale · 16-byte flits · Mesh XY and Mesh Bypass agree.

Performance numbers are gated by invariants

Gate	Accepted cases	Status
Legacy collective regression	11	PASS
Multicast correctness matrix	208	PASS
Tensor size/topology/root matrix	14	PASS
Tensor backpressure matrix	32	PASS
H100 broadcast / tensor / paired	30 / 15 / 2	PASS

Fail-fast contracts

- exact destination deliveries
- exact per-lane reduced values
- no duplicate / wrong-lane / incomplete request
- credit conservation and zero residual state
- provenance, trace hashes, and paired run inputs

Seven-stage full gate: 7 / 7 PASS

Retained artifacts: compact CSV/JSON snapshots, generated figures, run manifests, trace hashes, and exact reproduction commands.

What the evidence supports—and what it does not

Supported by the experiments

- tree replication removes duplicated Mesh link work
- distance-scaled diagonal shortcuts improve the tested sweep
- multi-flit tensor replay removes scalar request serialization
- all mechanisms satisfy their completion contracts

Not claimed

- area, energy, repeater, or timing-closure results
- NVLink/NVSwitch or end-to-end model speedup
- full training trace fidelity—the H100 source is a microbenchmark
- bounded hardware tensor tables or arithmetic pipeline latency

Important neutral result: the H100 collective placement selects zero express hops. Mesh XY and Mesh Bypass therefore match—adding links does not imply every workload uses them.

Two owners, shared integration

CHUNYU LIU

Multicast + bypass

Implementation, correctness checks, H100 trace capture, optimization, performance sweep, and analysis.

BOYAN PU

Tensor all-reduce

Datapath, replay contract and validator, backpressure stress tests, tensor statistics, and paired evaluation.

Integrate

merge mechanisms and runners



Review

cross-check invariants and regressions



Reproduce

rerun final snapshot and report

Both authors jointly performed integration, regression review, final reruns, and report verification.

CONCLUSION

Expose useful structure—then measure its cost

39.51%

less internal-link traffic with tree multicast

1.629×

latency speedup with distance-scaled diagonal bypass

4.728×

shorter H100 replay window with tensor streaming

Share

where common route prefixes remove real link work.

Shortcut

only when wire cost and traffic make the hop valuable.

Stream

with explicit request units and completion invariants.

Thank you · Questions?